AMERICAN OSTEOPATHIC ASSOCIATION

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March 14, 2025

Sethuraman Panchanathan
Director
National Science Foundation
2415 Eisenhower Ave
Alexandria, VA 22314

Re: Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Dear Director Panchanathan,

The American Osteopathic Association (AOA), on behalf of the more than 197,000 osteopathic physicians (DOs) and medical students we represent, appreciates this opportunity to provide the National Science Foundation (NSF) and White House Office of Science Technology Policy (OSTP) with information and policy recommendations to sustain and enhance artificial intelligence (AI) innovation in the United States. The AOA encourages OSTP to strengthen federal AI regulation and agency coordination in the healthcare space to increase adoption of innovative AI-enabled tools among physicians and ensure patient safety.

Innovation in AI-driven technologies holds great promise for improving patient care through promoting better outcomes, increasing efficiency, supporting physician decision-making, and enabling physicians to spend more time with patients. Osteopathic physicians, who are trained in patient-centered, whole-person care, play a critical role in fostering innovation, whether through development of new technologies, leading product adoption and implementation in their care settings, or using innovative care tools in their practice.

The AOA recognizes the importance of carefully balancing policies that ensure product safety and effectiveness while permitting developers sufficient flexibility to innovate and create unique products. Federal agencies must ensure oversight that provides clear and structured requirements around product performance transparency, safety, and effectiveness. In the absence of federal leadership on these issues and adequate oversight, states will adopt regulations, which would create a patchwork of regulations for developers to follow. Requiring developers to comply with 50 differing sets of AI-enabled device regulations would discourage innovation and increase costs on developers. Additionally, in order for broader adoption of novel technologies to take place, patients and physicians must be assured that products are safe and reliable. It is with this perspective that the AOA offers input on how federal agencies can support U.S. leadership in A.I. innovation and adoption in healthcare.

Ensuring Safety and Effectiveness of AI-Driven Products and Devices

The AOA urges the Food and Drug Administration (FDA) to use its full authority under medical device and other product approval pathways to ensure that AI-driven products are safe and effective. This includes clinical decision support tools, prescription digital therapeutics, remote monitoring products, risk management tools, as well as numerous other technologies that fall under FDA's definition of a software medical device. Currently, there are significant gaps in FDA's oversight, as well as in developers' transparency regarding their software's performance. Several studies have been published in recent years indicating that the current "risk based" regulatory framework for evaluating and approving AI devices may be inadequate, and that many products may not be sufficiently tested and validated. Although FDA has approved over 950 medical devices driven by AI, as many as 43% lack clinical

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validation data in their FDA submissions,[1] and at least 211 products have been recalled.[2] A stronger regulatory framework that supports positive patient outcomes is essential. FDA should pursue standardized regulatory mechanisms requiring transparency in AI device development, validation of datasets, and continuous monitoring of products post-approval to ensure ongoing safety, efficacy, and bias mitigation.

The FDA has begun to address these issues, and we strongly urge the Trump Administration to continue this work to provide confidence to patients, and to the physicians interacting with these products, that they work as intended. Most notably, FDA recently released guidance for maintaining safety and effectiveness in AI-devices at all stages of the product lifecycle. The guidance provides recommendations for applications under the 3 software device approval pathways and includes model submissions for developers. However, the guidance and corresponding regulations fall short in several key areas. FDA states in the guidance that the agency is concerned about erroneous outputs and bias with AI-enabled devices, but requirements for product summaries and device labeling are not sufficiently stringent to ensure that users and patients have necessary information on device development, testing, performance, and risks. End-users of AI-devices, including physicians and healthcare facilities, may be hesitant to purchase or adopt an AI-device if they do not know what data was used to train the device (e.g. the source of the data, whether it is real patient data or synthetic), or whether the product was validated via trials or other means. Additionally, as we describe in more detail below, physicians are required under Affordable Care Act Section 1557 regulation to assess the potential for biased outcomes when using software devices and are liable for such outcomes when using these products. Because physicians often do not have sufficient information on device training and performance, they are likely to be more hesitant to adopt new, innovative tools due to the current state of transparency and liability.

The AOA will be providing separate, more detailed comments to FDA regarding the AI-enabled device total product life cycle guidance and recommending that the administration finalize the document to provide clear instruction for developers and manufacturers. **As part of its efforts to develop on AI action plan, the AOA strongly recommends that the Office of Science and Technology Policy prioritize policies to increase data transparency regarding development of AI-enabled devices in healthcare, and strengthen FDA regulation governing approvals and labeling to encourage trust in the AI-enabled device development process.**

Similarly, many AI-driven tools used by enterprises in operations, but not used in direct patient care, lack appropriate oversight. However, these products can still result in erroneous or inappropriate outcomes that harm patient care. Such products include patient population management and prior authorization decision tools used by payers. **We strongly urge FDA to use existing authority to provide oversight to these products, and if it lacks authority, work with Congress to ensure appropriate oversight.**

We also wish to highlight that AOA is working closely with the Digital Medicine Society (DiMe) on an evaluation framework for AI-enabled products and has convened a panel of experts focused on these issues. These experts include developers, health system leaders, and leaders in medical education focused on innovation. **We welcome the opportunity to work more closely with the Administration on this effort and share technical input where helpful.**

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[1] Chouffani El Fassi, S., Abdullah, A., Fang, Y. et al. Not all AI health tools with regulatory authorization are clinically validated. Nat Med (2024).

[2] Muehlematter et al. "FDA-cleared artificial intelligence and machine learning-based medical devices and their 510(k) predicate networks." The Lancet Digital Health, Sep 2023

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Supporting Adoption through Appropriate Payment
To achieve wide adoption of innovative technologies across healthcare, physicians and enterprises must see return on investment for these products. How different AI-enabled software and devices are paid for varies by the nature of the product, and challenges exist across payment systems and product use cases. Without a strong payment framework, adoption of AI tools will occur inequitably across sites of care (e.g. physician practices vs. health systems), limiting access to innovative tools to sites of care that have more capital and can implement tools at scale.

Products that support operations, such as helping with population management, or automating documentation and billing (e.g. ambient AI) are not separately billable under payment systems. Under the physician fee schedule specifically, these are considered practice expense, and CMS' current PE methodology does not fully account for these costs. Additionally, due to the slim margins that many physician practices operate under, practices often don't have sufficient capital to invest in these products even though they may improve patient care, enable them to spend more time at the bedside with patients, and help them better target practice resources. In addition to improving patient care, these tools can help address physician burnout, drive efficiency, and improve physicians' experience in delivering care.[3]

Tools that function as "software as a service" that are used in direct patient care and have an associated billable service can be paid directly under various payment systems. However, there are significant flaws in how payment is calculated which is driving inequitable adoption across sites. While hospitals receive separate payment for many of these tools, physicians under the physician fee schedule do not. Under the hospital inpatient prospective payment system, manufacturers can apply for a new technology add-on payment until payment for the new technology can be bundled in the payment rates for applicable Medicare severity-diagnosis related groups. In the hospital outpatient prospective payment system (OPPS), software as a service is often classified as a separately payable service, not ancillary or supportive to the bundled service the software is enabling. Under the physician fee schedule, CMS has not created national payment rates for most software, and payment is "carrier priced", resulting in Medicare Administrative Contractors to determine payment on a case-by-case basis. However, MACs lack sufficient data on product costs and often establish inadequate payment rates.

Prescription digital therapeutic products are generally those that are approved by the FDA to be prescribed by a physician to manage or treat an injury or disease. They are typically administered by patients themselves on a phone, tablet, smartwatch, or similar device, and they primarily use software to diagnose or treat an illness or injury. These devices do not fall into a defined Medicare benefit category and lack a clear payment mechanism, even though they may benefit patient care and improve outcomes.

A recent report by the Medicare Payment Advisory Commission highlights many of the issues described above.[4] While AOA's concerns primarily focus on Medicare, state coverage of these technologies under Medicaid programs is patchwork and often limited.

It is also important to note that physician services associated with "software as a service" cannot be leveraged for preventive care. This is because many services, such as remote monitoring, require specific diagnoses in order to be

[3] Michael Albrecht, Denton Shanks, Tina Shah, Taina Hudson, Jeffrey Thompson, Tanya Filardi, Kelli Wright, Gregory A Ator, Timothy Ryan Smith, Enhancing clinical documentation with ambient artificial intelligence: a quality improvement survey assessing clinician perspectives on work burden, burnout, and job satisfaction, JAMIA Open, Volume 8, Issue 1, February 2025
[4] MedPAC June 2024 Report, Chapter 4. Available here.

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paid. We urge CMS to identify use cases where it would be appropriate to pay for such services for prevention. CMS could also consider leveraging its authority under the Center for Medicare and Medicaid Innovation (CMMI) to establish pilot programs or demonstration projects to incentivize the integration and evaluation of AI technologies into practices, specifically for preventive healthcare services. These pilots could evaluate cost-effectiveness, patient outcomes, and operational efficiencies that could guide broader AI payment reforms.

Overall, the AOA recommends that the Centers for Medicare & Medicaid Services (CMS) prioritize policy development to improve payment for innovative technologies by:

- Identifying ways to help practices invest in products that are considered operational or "practice expense" that may improve patient care and enable physicians to spend more time with patients;
- Develop a comprehensive approach to payment of "software as a service" under the Medicare physician fee schedule and establish payment for use of such technology in preventive care;
- Develop a payment pathway for digital therapeutics; and
- Work with congress in areas where its authority to establish payment is limited.

Ensuring adequate payment and the ability of practices to invest in new technologies is central to adoption. Developing comprehensive payment reform will ensure that the U.S. healthcare system remains at the cutting edge of implementing innovative tools in patient care, improving outcomes across the country.

Addressing Liability Concerns

The AOA is concerned about the liability currently placed on physicians for potential bias resulting from use of AI-enabled devices. In 2024, HHS Office of Civil Rights (OCR) modified regulations for section 1557 of the Affordable Care Act, requiring physicians and other provider entities to monitor how point-of care decision support tools perform across different populations. It also makes them responsible when decisions using these products result in biased outcomes. If the AI software is trained with data that measures race, sex, age, disability, or other demographic factors, physicians are required to make "reasonable efforts" to mitigate risk of discrimination when the tool is used for health programs or activities. The AOA firmly believes that bias and discrimination should be prevented in the development and approval process to ensure patient safety and device effectiveness.

However, the data used to train software for these patient care decision support tools is not easily available to physicians, nor is data on how many devices perform across populations made available. Developers are not always required to disclose data validation sources or performance during the development and FDA clearance stages. As a result, physicians are placed in a difficult position where they have an obligation to reduce discrimination risk but do not have the information available to know how a device performs and how to mitigate risk. In a 2024 survey, 88 percent of surveyed physicians stated physician liability for errors in AI models would impact adoption of AI tools into their practice[5]. As an example of a challenge physicians face, many tools are not validated in a local site and may not be generalizable. A tool that was trained for a large academic health system, may not be as useful in a rural health population. However, current law may place an unfair liability burden on physicians.

[5] AMA Physician AI Sentiment Report. 2024. Available here.

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Further, physicians train to practice medicine, not to moderate software development and oversight. Physicians without a data analysis background may not have the breadth of knowledge required to determine which data sets are appropriate for certain devices. Current liability frameworks already hold physicians accountable when they make medical errors based on known software errors or performance issues with devices, but physicians should not be held accountable for errors resulting from device development flaws they cannot reasonably be aware of. This inappropriate shifting of liability from developers to providers makes physicians hesitant to be early adopters of novel tools and slows delivery of innovative care.

The burden of training and maintaining a safe product should reside with the developer and manufacturers, not the end-users. **OCR should revise 45 CFR 92.210 to remove the risk identification burden from physicians to encourage physician uptake of AI-enabled software without fear of liability for products developed and trained outside of their control**.

Ensuring Consistent Standards and Cross-Agency Coordination

An interoperable data ecosystem that facilitates sharing of patient data across sites of care is essential to care coordination, software adoption, and innovation. We urge the OSTP to work closely with the HHS Assistant Secretary for Technology Policy (ASTP) and support the office's work to this end. Additionally, as many of the issues outlined above overlap across the functions of different agencies (e.g. ensuring transparency, mitigating unintended outcomes, and addressing liability), it is essential that agency efforts be well coordinated, and ASTP can play a critical role in cross-agency coordination.

Conclusion

The AOA is pleased to have provided comments and recommendations for this request for information. We look forward to continuing to work with OSTP and NSF on developing policy recommendations for the Artificial Intelligence Action Plan. Should you have any questions regarding our comments or recommendations, please contact John-Michael Villarama, Vice President for Public Policy at [redacted] at any time should we be able to support your efforts.

Sincerely,

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Chief Executive Officer, AOA

Teresa A. Hubka, DO, FACOOG (dist.)
President, AOA